

Division by t

1. Find the Laplace transform of $\frac{1-e^{at}}{t}$

ans: $\log\left(\frac{s-a}{s}\right)$

2. Find the Laplace transform of $\frac{e^{at} - \cos bt}{t}$

ans: $\log\left(\frac{\sqrt{s^2+b^2}}{s-a}\right)$

3. Find the Laplace transform of $\frac{e^{-at} - e^{-bt}}{t}$

ans: $\log\left(\frac{s+b}{s+a}\right)$